



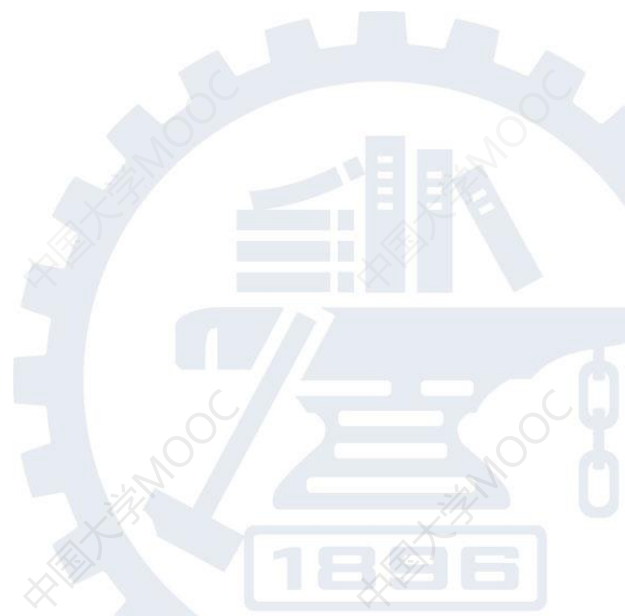
北京交通大学
BEIJING JIAOTONG UNIVERSITY



数学物理方程引言

郑神州

北京交通大学 理学院数学系





一、应用偏微分方程技术

- 物理、数学及工程问题，涉及**数学物理方程模型**
时-空变化的数学描述
- 应用范围：热力学、电磁学、流体力学、弹性力学、理论力学、相对论、量子力学，拓展到生物学、**金融数学**等
- 用到知识：数学分析、函数论、常微分方程、代数、泛函分析和计算机技术。



为什么学这课程?

- 世界是多元的、变化的,万物都随时-空演化. 如何科学、定量地地刻划,数学物理方程就是科学定量地研究多元问题的入门课程。
- 科学研究两大任务:一是从已知量测量数据推理无法企及的未知量(PDE边值问题),二是从现有状态预测未来演化规律(始值问题、混合问题)。
- 要明白流体、电磁学、天体演化、相对论、量子科学等物理学、要懂得世界万物演化,从这里出发。
- 科学、严谨性:数学是科学研究的共同语言,从物理原理出发,用数学的严谨演绎出实际问题的结论,它是物理规律的数学描述。



- **对象:**理科生为主, 如: 数学与应用数学、信息与计算科学、理科试验班等专业
- **学时:**教学安排32学时
- **先修课程:**数学分析、常微分方程、高等代数、复变函数



章节安排:

课程主旨: 训练物理和数学中三类典型方程定解问题的求解和适定性分析:

- 第一章 偏微分方程导出与定解问题
- 第二章 波动方程
- 第三章 热传导方程
- 第四章 热传导方程



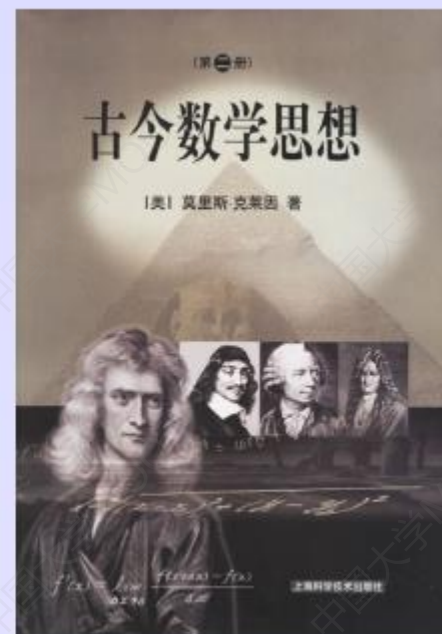
具体内容安排:

- 数学物理方程**模型建立**(三类典型方程和定解条件的导出), 有关的**基本概念**。
- 展开对三类典型方程在不同定解条件下的**求解方法**: 特征线法(行波法)、齐次化原理、对称延拓法、分离变量法(特征函数展开法)、积分变换法、广义函数、基本解、Green函数法等
- 以数学物理方程**适定性**为主线, 讨论解存在性, 以及**唯一性和稳定性方法**: 能量不等式估计、Gronwall不等式、极值原理以及最大模估计。



北京交通大学
BEIJING JIAOTONG UNIVERSITY

偏微分方程的整体介绍





- 从事偏微分方程理论及其应用研究的人数众多：
据说占有从事数学及其应用研究的人数的一半以上。



北京交通大学

千禧年大奖难题(Millennium Prize Problems)

克雷数学研究所(Clay Mathematics Institute) 于2000年5月24日

- P/NP问题(P versus NP)
- 霍奇猜想(Hodge Conjecture)
- 黎曼猜想(Riemann Hypothesis)
- 庞加莱猜想(Poincare Conjecture): 格里戈里·佩雷尔曼(2006)
- 杨-米尔斯存在性与质量间隙(Yang-Mills Existence and Mass Gap)
- 纳维-斯托克斯存在性与光滑性(Navier-Stokes existence and smoothness)
- 贝赫和斯维讷通-戴尔猜想(Birch and Swinnerton-Dyer Conjecture)





纳维-斯托克斯存在性与光滑性(Navier-Stokes existence and smoothness)



Incompressible Navier-Stokes equations

$$\begin{aligned} \text{(NS)} \quad & \partial_t v - \nu \Delta v + (v \cdot \nabla)v + \nabla p = f \\ & \operatorname{div} v = 0 \end{aligned} \quad \text{in } \Omega \times (0, T).$$

Here f is a given body force and $\nu > 0$ is the viscosity constant. We denote

$$(1.1) \quad \partial_t = \frac{\partial}{\partial t}, \quad \partial_i = \frac{\partial}{\partial x_i},$$

and

$$(1.2) \quad (v \cdot \nabla)v = v_j \partial_j v = \partial_j (v_j v).$$



- Equation of continuity (conservation of mass).

$$(1.6) \quad \frac{d}{dt} \int_E \rho dx = - \int_{\partial E} \rho v \cdot \mathbf{n} dS.$$

- Equation of motion (balance of momentum).

$$(1.8) \quad \frac{d}{dt} \int_E \rho v_i dx = - \int_{\partial E} \rho v_i v \cdot \mathbf{n} dS + \int_E f_i dx + \int_{\partial E} \sigma_{ij} \mathbf{n}_j dS,$$

- Equation of state

➤ Incompressible

$$\partial_j \sigma_{ij} = \nu \Delta v_i - \partial_i p.$$

➤ compressible

$$\partial_t(\rho v_i) + \operatorname{div}(\rho v_i v) = f_i - \partial_i \rho^\alpha + \nu \Delta v_i + \left(1 - \frac{2}{n}\right) \nu \partial_i \operatorname{div} v.$$



不可压两相流动

具有清晰界面的气液/液液两相流动广泛存在于自然界和工程应用领域





铁磁流体

铁磁流体(ferrofluid)是一种由纳米级的磁性固体颗粒、基载液以及表面活性剂三者混合而成的稳定的胶状液体,其既具有液体的流动性又具有固体磁性材料的磁性。

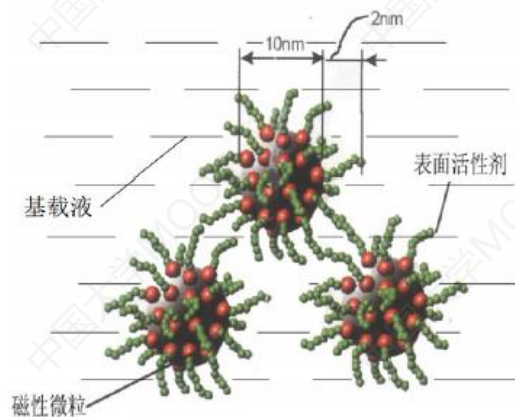


图 铁磁流体微观结构

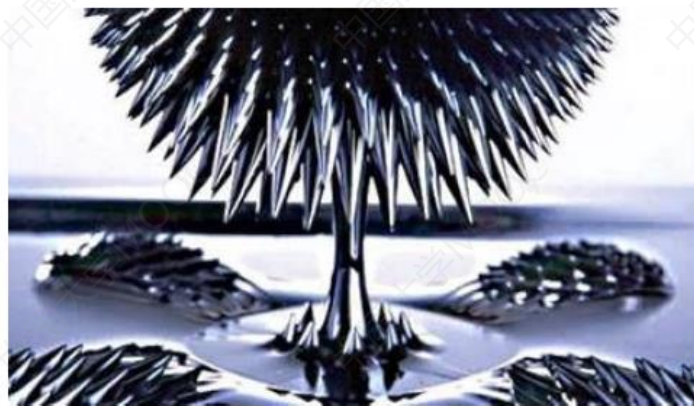


图 磁场作用下的铁磁流体

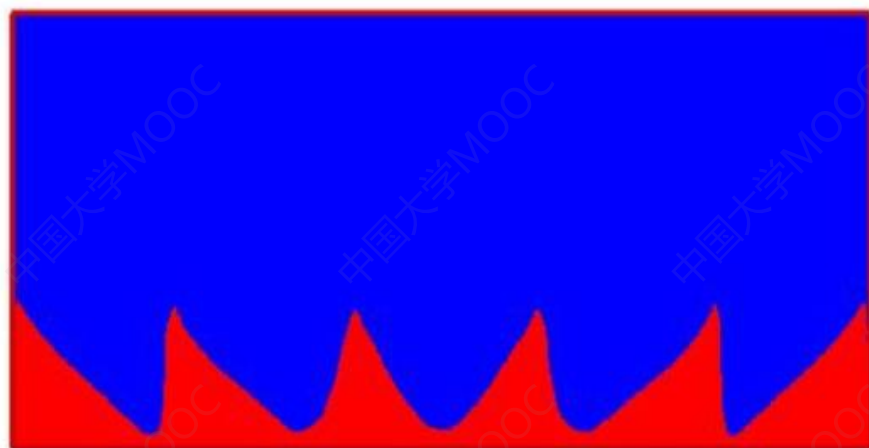


图 非均匀磁场下的“刺猬”效应



铁磁流体动力学

- Maxwell 方程

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{H} = 0$$

- 本构关系

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = \mu_0(1 + \chi)\mathbf{H}$$

$$\chi = \frac{M_s}{H} \left[\coth\left(\frac{3\chi_0 H}{M}\right) - \frac{M}{3\chi_0 H} \right]$$

- 磁势方程

$$\mathbf{H} = -\nabla\psi$$

$$\nabla \cdot (\mu \nabla\psi) = 0$$

- Navier-Stokes 方程

$$\nabla \cdot \mathbf{u} = 0$$

$$\rho \left[\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) \right] = -\nabla p + \eta \nabla^2 \mathbf{u} \\ + \nabla \cdot \boldsymbol{\tau}_m + \mathbf{f}_s + \mathbf{f}_b$$

- Maxwell 应力

$$\boldsymbol{\tau}_m = -\frac{\mu_0}{2} |\mathbf{H}|^2 \mathbf{I} + \mathbf{H}\mathbf{B}$$

$$\mathbf{f}_m = -\frac{\mu_0}{2} \nabla(|\mathbf{H}|^2) + (\nabla \cdot \mu \mathbf{H})\mathbf{H} + (\mu \mathbf{H} \cdot \nabla)\mathbf{H}$$

$$= -\frac{\mu_0}{2} \nabla(|\mathbf{H}|^2) + (\mu \mathbf{H} \cdot \nabla)\mathbf{H}$$

$$= -\frac{\mu_0}{2} \nabla(|\mathbf{H}|^2) + \mu \left[\frac{1}{2} \nabla(\mathbf{H} \cdot \mathbf{H}) - \mathbf{H} \times (\nabla \times \mathbf{H}) \right]$$

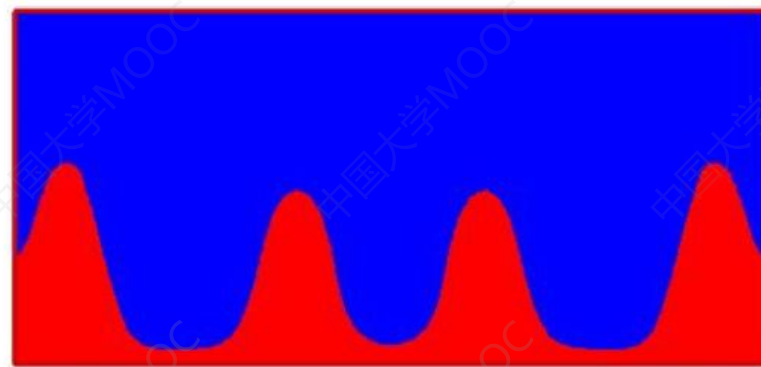
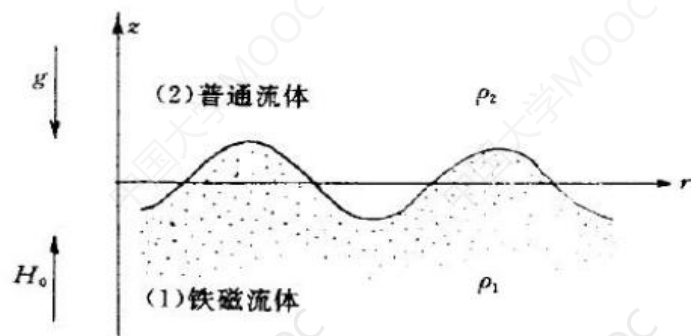
$$= \frac{\mu - \mu_0}{2} \nabla(|\mathbf{H}|^2) = \frac{\mu_0 \chi}{2} \nabla(|\mathbf{H}|^2)$$



铁磁流体界面动力学

Normal field(Rosensweig)不稳定性

$$l_c = 2\pi \sqrt{\frac{\sigma}{g\Delta\rho}}$$
$$M_c = \frac{2(\mu_1 + \mu_2)}{\mu_1\mu_2} \sqrt{\Delta\rho g \sigma}$$





1. 一阶守恒律方程

⊙ Euler方程（激波、涡流、湍流等）



Bow Shock generated by a Solar Explosion (太阳爆炸产生的弓形波)

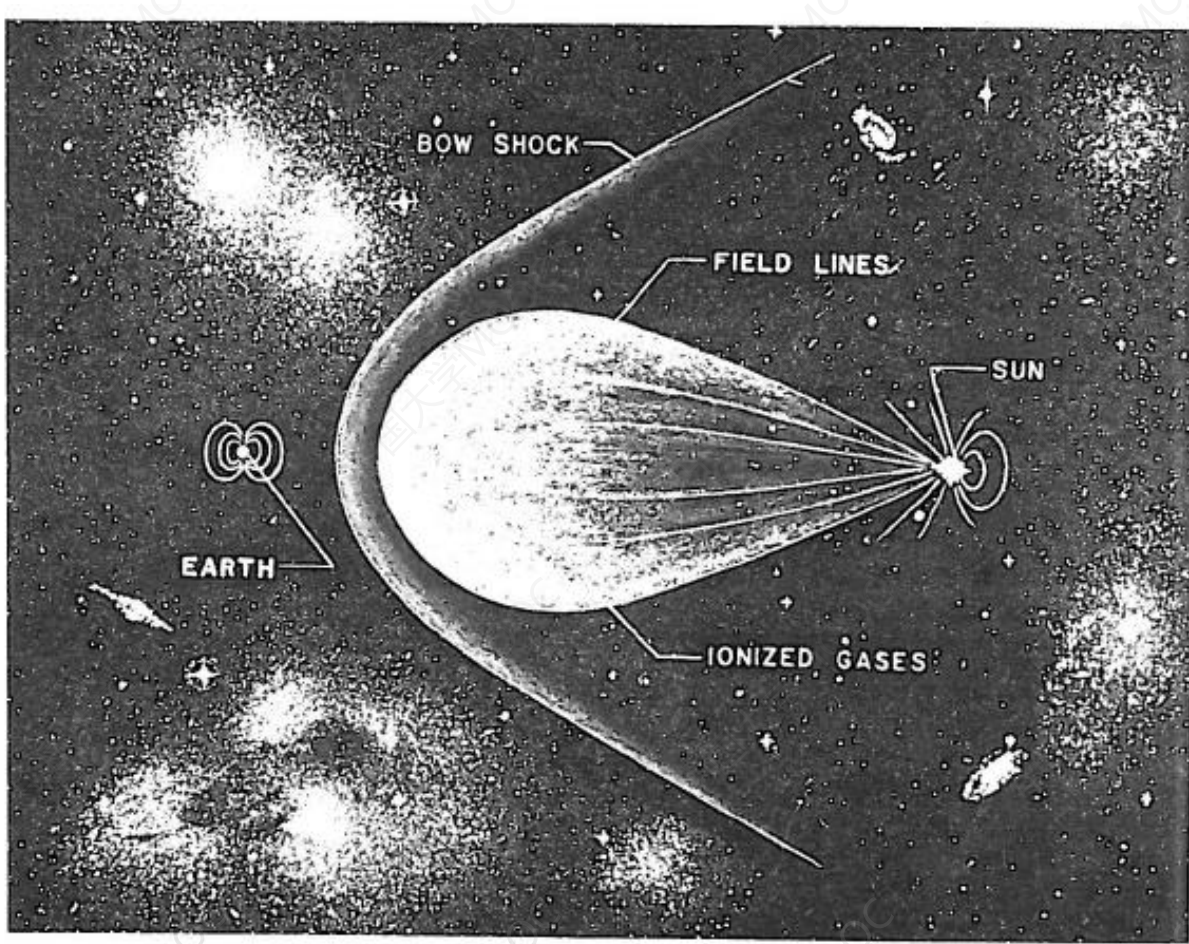


FIG. 50: SOLAR EXPLOSION



北京交通大学
BEIJING JIAOTONG UNIVERSITY

Shock Waves by Blunt-Nosed Supersonic Aircrafts (钝型航空器激波)

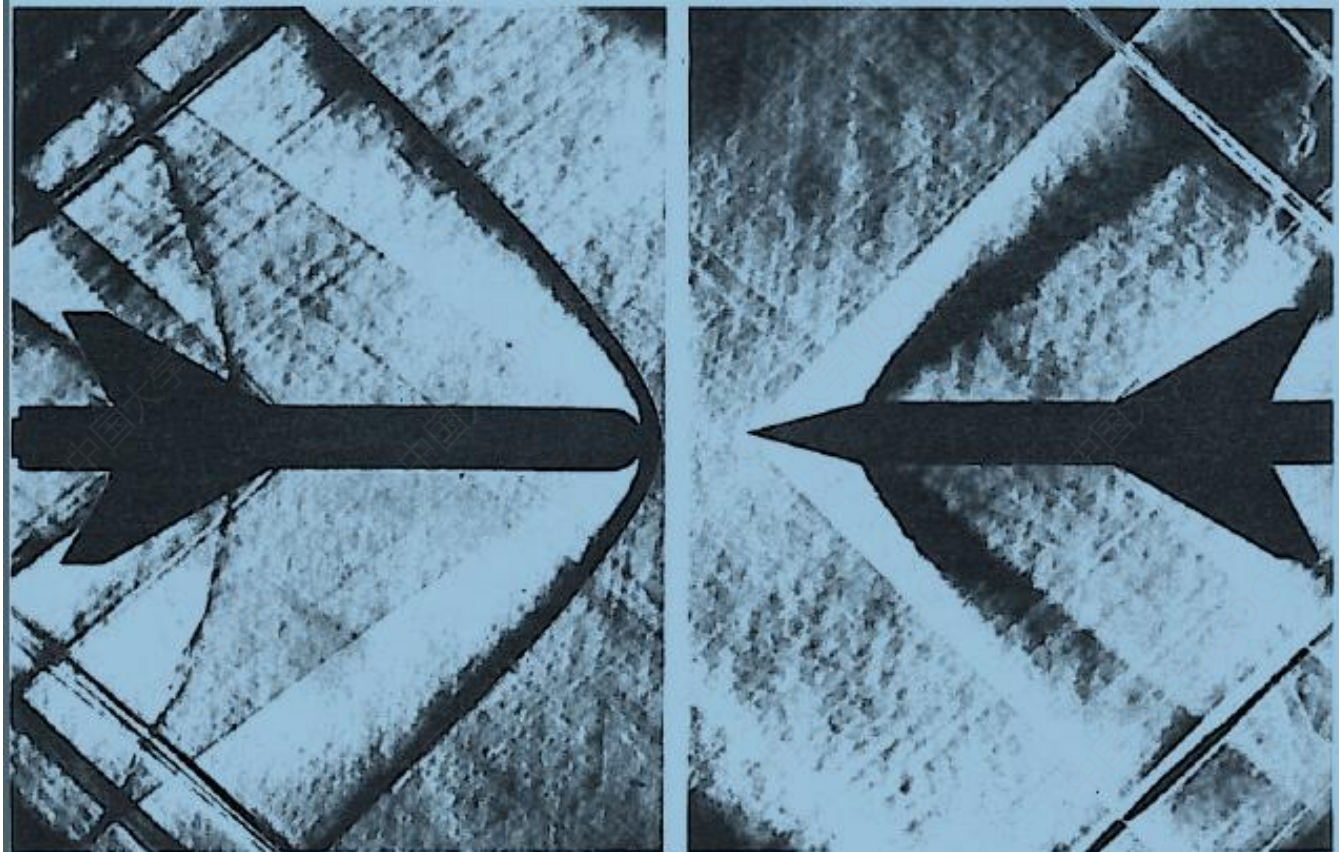


FIG. 41: SHOCK WAVES ABOUT MODEL AEROSPACECRAFT



北京交通大学
BEIJING JIAOTONG UNIVERSITY

Shock Waves Breaking the Sound Barrier (跨音障激波)





北京交通大学
BEIJING JIAOTONG UNIVERSITY

Blast Wave from TNT Surface Explosion (TNT爆炸的表面中激波)

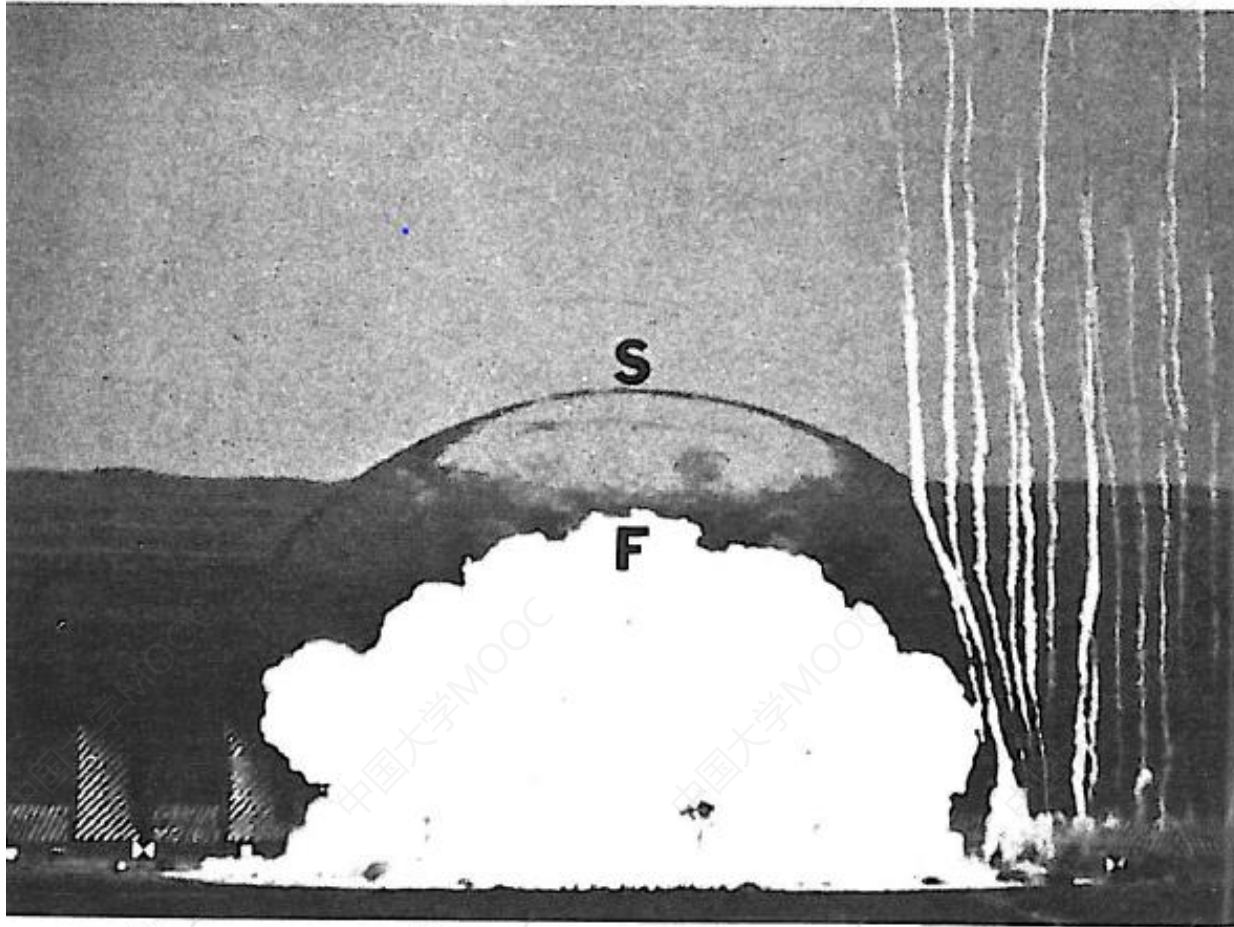
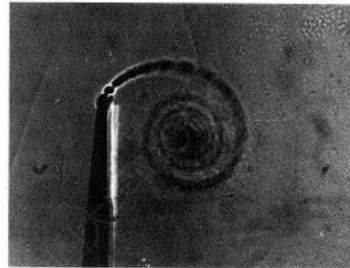


FIG. 22: EXPLOSION FROM A 20-TON HEMISPHERE OF TNT



Vortex from a Wedge



82. Vortex from a wedge in a shock tube. This schlieren photograph shows the vortex that spirals from the tip of a thin wedge after the air is set in motion normal to it by the passage of a weak plane shock wave, which is out of sight to the right. Other photographs show that the flow pattern is "conical" or "pseudo-stationary," remaining always similar to itself but growing in size in proportion to the time. Photograph by Walker Blackney





可压缩Euler方程 (质量、动量、能量守恒)

Full Euler Equations for Compressible Fluids (E-1)

$$\begin{cases} \partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0, \\ \partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v}) + \nabla p = 0, \\ \partial_t (\rho E) + \nabla \cdot (\rho \mathbf{v} (E + \frac{p}{\rho})) = 0 \end{cases} \quad (t, \mathbf{x}) \in \mathbb{R}_+^{d+1} := (0, \infty) \times \mathbb{R}^d$$

Constitutive Relations: $p = p(\rho, e), \quad E = \frac{1}{2}|\mathbf{v}|^2 + e$

$\tau = \frac{1}{\rho}$ — Deformation gradient (specific volume for fluids, strain for solids)

$\mathbf{v} = (v_1, \dots, v_d)$ — Fluid velocity with $\mathbf{m} = \rho \mathbf{v}$ the momentum vector

p — Scalar pressure

E — Total energy with e the internal energy which is a given function of (τ, p) or (ρ, p) defined through thermodynamical relations



北京交通大学
BEIJING JIAOTONG UNIVERSITY

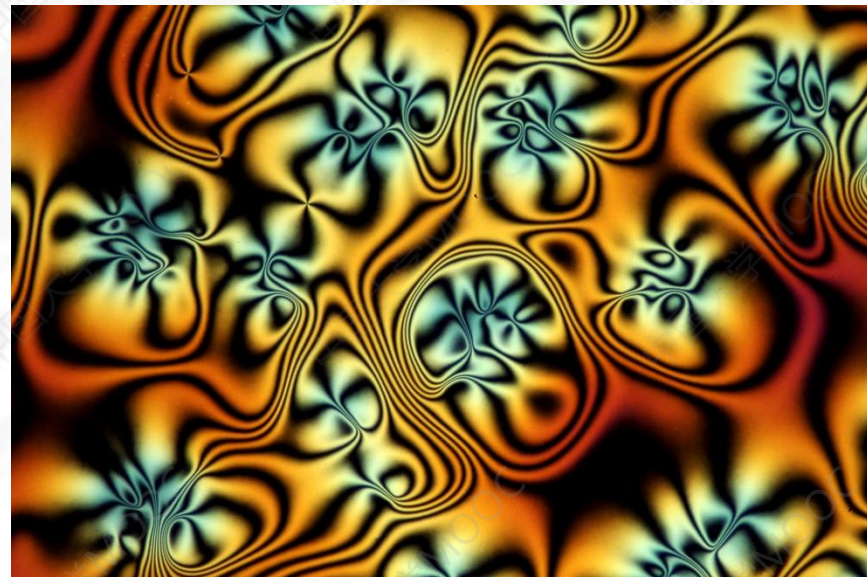
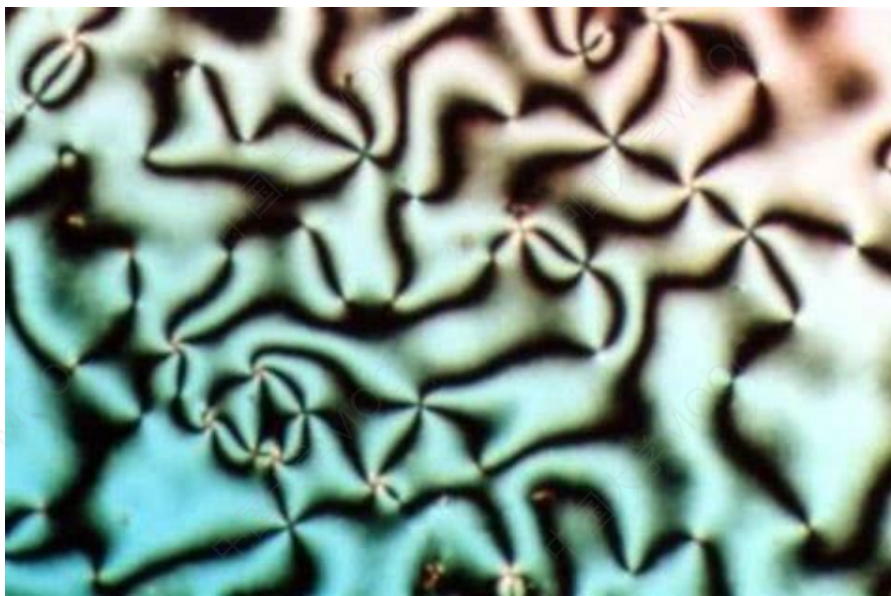
2. Turbulence (湍流)





北京交通大学
BEIJING JIAOTONG UNIVERSITY

3. Liquid Crystal (液晶)





free-energy functional

$$I_{OF}(\mathbf{n}) = \int_{\Omega} W(\mathbf{n}, \nabla \mathbf{n}) d\mathbf{x},$$

where

$$W(\mathbf{n}, \nabla \mathbf{n}) = K_1(\operatorname{div} \mathbf{n})^2 + K_2(\mathbf{n} \cdot \operatorname{curl} \mathbf{n} + q_0)^2 + K_3|\mathbf{n} \times \operatorname{curl} \mathbf{n}|^2 \\ + (K_2 + K_4)(\operatorname{tr}(\nabla \mathbf{n})^2 - (\operatorname{div} \mathbf{n})^2),$$

and the coefficients K_i, q_0 are constants, with $q_0 = 0$ for nematics and $q_0 \neq 0$ for cholesterics. (The constants K_i, q_0 , together with other material constants specified below, depend on temperature, but for simplicity we do not make this explicit and assume we are working at a constant temperature at which the liquid crystal is in the nematic phase.)



free-energy functional (液晶数学模型2)

$$I_{LdG}(\mathbf{Q}) = \int_{\Omega} \psi(\mathbf{Q}, \nabla \mathbf{Q}) d\mathbf{x}. \quad (2.3)$$

The free-energy density can be written as $\psi(\mathbf{Q}, \nabla \mathbf{Q}) = \psi_B(\mathbf{Q}) + \psi_E(\mathbf{Q}, \nabla \mathbf{Q})$, where $\psi_B(\mathbf{Q}) = \psi(\mathbf{Q}, 0)$. The bulk energy ψ_B is by frame-indifference an isotropic function of $\mathbf{Q}$, so that

$$\psi_B(\mathbf{Q}) = \hat{\psi}_B(\text{tr } \mathbf{Q}^2, \det \mathbf{Q}) \quad (2.4)$$

the elastic energy is

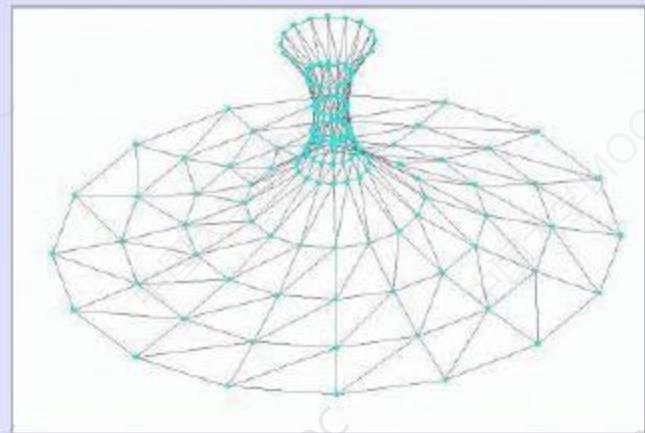
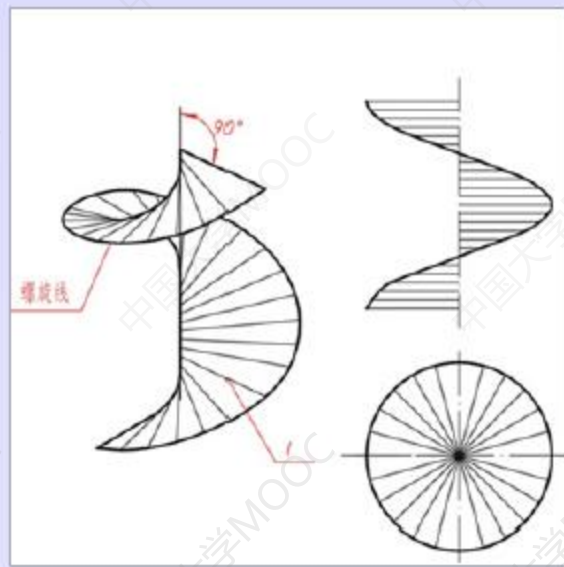
$$\begin{aligned} \psi_E(\mathbf{Q}, \nabla \mathbf{Q}) &= \sum_{i=1}^5 L_i I_i, \\ I_1 &= Q_{ij,j} Q_{ik,k}, \quad I_2 = Q_{ik,j} Q_{ij,k}, \\ I_3 &= Q_{ij,k} Q_{ij,k}, \quad I_4 = Q_{lk} Q_{ij,l} Q_{ij,k}, \quad I_5 = \varepsilon_{ijk} Q_{il} Q_{jl,k}, \end{aligned}$$



4. 极小曲面方程

$$\frac{\partial}{\partial x} \left(\frac{u_x}{\sqrt{1 + u_x^2 + u_y^2}} \right) + \frac{\partial}{\partial y} \left(\frac{u_y}{\sqrt{1 + u_x^2 + u_y^2}} \right) = 0,$$

$$(1 + u_y^2)u_{xx} - 2u_xu_yu_{xy} + (1 + u_x^2)u_{yy} = 0.$$





5. 量子力学方程

- 光的波动、粒子两重性
- 波函数的概率解释

Schrödinger flows

The unique solution $u(t, x) \in C^1(\mathbb{R}; L^2(\mathbb{R}^d))$ to

$$\begin{cases} \partial_t u = i\Delta u \\ u(0) = f \in L^2(\mathbb{R}^d) \end{cases}$$

is given by

$$e^{it\Delta} f(x) := (2\pi it)^{-\frac{d}{2}} e^{i\frac{|x|^2}{4t}} * f(x) = \frac{e^{i\frac{|x|^2}{4t}}}{(2\pi it)^{\frac{d}{2}}} \int_{\mathbb{R}^d} e^{-i\frac{x \cdot y}{2t}} e^{i\frac{|y|^2}{4t}} f(y) dy.$$



6、不确定性原理

定理 2-1. 假定 $f \in L^2(R)$, 如果我们有

$$\int_R |\hat{f}(s)|^2 ds = \int_R |f(x)|^2 dx = 1$$

对所有 $x, s \in R$ 都成立, 那么则可以推出下列不等式

$$\left(\int_R |x|^2 |f(x)|^2 dx \right) \cdot \left(\int_R |s|^2 |\hat{f}(s)|^2 ds \right) \geq \frac{1}{(4\pi)^2}$$

成立, 从而迫使 f 和 $\hat{f}$ 中的至少一个不被过于集中在原点附近。

Heisenberg: “The more precisely the position is determined, the less precisely the momentum is known in this instant, and vice versa.”



Hardy Uncertainty Principle

Theorem (Hardy Uncertainty Principle (H.U.P.))

Let $f : \mathbb{R} \rightarrow \mathbb{C}$ be (a measurable function) such that

$$|f(x)| \leq C_1 e^{-\pi a x^2}, \quad |\widehat{f}(\xi)| \leq C_2 e^{-\frac{\pi \xi^2}{a}},$$

for some $a, C_1, C_2 > 0$. Then $f(x) = \gamma e^{-\pi a x^2}$, for some $\gamma \in \mathbb{C}$.



Unique continuation properties for Schrödinger Eqns.

The previous results read as follows

Theorem (U.C.P. for Schrödinger - L^∞ -version)

Let $f \in L^2(\mathbb{R}^d)$ be such that

$$|f(x)| \leq C_1 e^{-|x|^2/\beta^2}, \quad \left| e^{iT\Delta} f(x) \right| \leq C_2 e^{-|x|^2/\alpha^2},$$

for some $\alpha, \beta, C_1, C_2, T > 0$. Then,

$$\alpha\beta < 4T \Rightarrow f \equiv 0$$

$$\alpha\beta = 4T \Rightarrow f \text{ is a constant multiple of } e^{-\left(\frac{1}{\beta^2} + \frac{i}{4T}\right)|x|^2}.$$



7. Schrödinger Equation

Schrödinger Equation with a Potential

$$u = u(x, t)$$

$$\partial_t u = i(\Delta + V)u \quad x \in \mathbb{R}^n$$

Non-Linear Schrödinger Equation

$$\partial_t u = i(\Delta u + F(u, \bar{u})) \quad x \in \mathbb{R}^n \quad t \in [0, T].$$



北京交通大学
BEIJING JIAOTONG UNIVERSITY



THANK YOU FOR YOUR ATTENTION

